

Network Perimeter Hardening & Behavioral Threat Hunting

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1. Introduction

The purpose of this project was to understand how network hardening and threat hunting are performed in a practical environment. I built a small virtual lab consisting of multiple machines and used different security tools to observe normal traffic, perform reconnaissance, reduce unnecessary exposure, and implement firewall rules based on the principle of least privilege.

This project helped me gain hands-on experience in identifying network services, analyzing logs, and understanding how attackers and defenders interact within a network.

2. Lab Architecture

The lab environment consisted of the following systems:

Machine	IP Address	Role
pfSense	10.0.0.1	Firewall
Wazuh	10.0.0.10	SIEM Platform
Ubuntu 16	10.0.0.20	Web Server
Windows 7	10.0.0.30	Client Machine
Kali Linux	10.0.0.40	Attacker Machine

The pfSense firewall-controlled communication between the systems, Wazuh collected security logs, Ubuntu hosted the web services, Windows acted as a normal user machine, and Kali Linux was used for reconnaissance and testing activities.

3. Establishing a Traffic Baseline

I started the project by capturing normal network traffic to understand what regular activity looks like inside the environment. This was important because knowing the baseline makes it easier to identify suspicious behavior later.

I observed three common types of traffic:

- ICMP traffic generated through ping requests
- SSH traffic during remote login sessions
- HTTP traffic while accessing the Apache web server

Using Wireshark, I analyzed packet details and became familiar with the normal communication patterns between the systems.

4. Attack Surface Reduction

Before making any changes, I checked which services were running on the Ubuntu server using the `ss -tulnp` command. This helped me identify open ports and understand the current attack surface of the machine. After reviewing the running services, I disabled unnecessary services that were not required for the project. I then performed another scan to verify that the exposed ports had been reduced, this process showed me how reducing unnecessary services can significantly decrease the number of possible entry points that an attacker could target.

5. Firewall Hardening

I configured firewall rules in pfSense using the principle of least privilege, allowing only the traffic that was necessary for the lab environment. Instead of permitting unrestricted communication, I created rules that limited access to required services only. Implementing these rules helped me understand how proper firewall configurations can strengthen network security and reduce unnecessary exposure.

6. Reconnaissance Findings

Using Kali Linux, I performed several Nmap scans, including host discovery, service detection, and full port scans.

From these scans, I observed:

- The Ubuntu server was running only the services that were required after hardening.
- Disabled services no longer appeared in scan results.
- Firewall restrictions affected which ports and hosts could be reached.
- Reconnaissance activities generated noticeable events in logs and monitoring tools.

By comparing the scan results before and after hardening, I could clearly see the impact of security improvements on the network.

7. Conclusion

This project helped me understand the importance of establishing normal traffic baselines before investigating potential threats. I learned how reducing unnecessary services and applying least-privilege firewall rules can improve overall network security. Working with tools such as Wireshark, Nmap, pfSense, and Wazuh also gave me practical experience in network monitoring, hardening, and basic threat hunting techniques.